

SMART INDIA HACKATHON 2026

An offline workstation that turns raw Bitcoin traffic into ranked, explainable investigative leads.

Problem Statement ID **SIH26146**

Theme **Cryptocurrency · Blockchain & Cybersecurity**

PS Category **Software**

Organisation **National Technical Research Organisation (NTRO)**

Team ID **SH-2026-VORTEX** · Team Name **Team Vortex** · 6 members

Prototype status **working, offline, measured — not a mockup**



NETRA

Network-Enhanced
Transaction Risk Analysis

AI-Powered Monitoring & Analysis
of Bitcoin Transaction Traffic

IDEA TITLE



TEAM VORTEX

One workstation. Two raw feeds in, ranked and explained leads out — and it never touches a network.

NETRA
RISK ANALYSIS

WORKSPACE

Overview

Queue

Case board

Operations map

Events & alerts

EVIDENCE

Campaign view

Canary feed

Reports library

INTELLIGENCE

Ingest & sources

Model scorecard

Threshold simulator

Dispositions

GOVERN

Audit log

Administration

Documents

Presentation tour

Administrator
admin · signed in

Queue

47 leads in this capture, ranked by the model's probability. 47 shown.

ExportAnalystOffline · 0 external callsExplainOffLightDark

Band ▼Search group, label or finding... 47 of 47 leadsTies are shown as ties: max means the forest was unanimous. The anomaly score never orders this list.

#	PRIORITY	BAND	STRONGEST FINDING - GROUP	CONF.	VALUE	COUNTRY
1	100	critical	Multi-country control NTR-1336 Touched a mixer Exchange-like landmark	max	910.10 BTC	IR TR
2	100	critical	Multi-country control NTR-2248 Touched a mixer Exchange-like landmark	max	815.84 BTC	IR NG
3	100	critical	Multi-country control NTR-4218 Exchange-like landmark	0.999	759.60 BTC	IR TR
4	100	critical	Multi-country control NTR-0191 Exchange-like landmark	0.999	624.81 BTC	IR PA
5	100	critical	Multi-country control NTR-9707 Exchange-like landmark	0.999	784.02 BTC	IR TR
6	100	critical	Multi-country control NTR-2351 Exchange-like landmark	0.999	953.67 BTC	HK NG
7	100	critical	Multi-country control NTR-8878 Exchange-like landmark	0.998	918.80 BTC	HK NG
8	100	critical	Multi-country control NTR-6734 Exchange-like landmark	0.998	717.15 BTC	IR RU
9	100	critical	Multi-country control NTR-7687 Exchange-like landmark	0.997	824.02 BTC	HK PA
10	100	critical	Multi-country control NTR-1293 Exchange-like landmark	0.996	833.64 BTC	NG RU
11	100	critical	Multi-country control NTR-0518 Touched a mixer Exchange-like landmark	0.996	1,016.57 BTC	PA RU
12	100	critical	Multi-country control NTR-1119 Exchange-like landmark	0.995	993.02 BTC	IR TR
13	99	critical	Multi-country control NTR-0566 Exchange-like landmark	0.994	1,012.67 BTC	NG PA

The queue, on the demonstration capture: 47 leads ranked by the model's own probability, each with the finding that raised it.

WHAT MAKES NETRA DIFFERENT

Correlates the two layers.

Most tools do blockchain OR network. NETRA fuses IP/port/timing with wallet/txid/amount, so a wallet group gets the country and hosting operator that sent it.

Explains every score.

Saabas decision-path attribution where base + listed + smaller factors equal the score shown, exactly — an investigator can check it.

Proves itself on planted ground truth.

98% ROC-AUC, 100% precision in the top-K, 86% recall — measured, not claimed.

Hunts campaigns, not just wallets.

Address poisoning and pool sweeps are cross-wallet patterns: one attacker, many victims.

Air-gapped by construction.

Zero external requests. Verified by reading every shipped file, not by promise.

Evidence-grade records.

Append-only audit log, cases, dispositions, model registry with rollback — chain of custody on the workstation.

TECHNICAL APPROACH



TEAM VORTEX

Six stages, one process, no server to call: ingest → identify → correlate → features → models → leads.

1. INGEST

CSV · JSON · XML · SQLite
4 readers, ~90 header aliases,
quality gate with a reason
for every row set aside

2. IDENTIFY

NTR-#### wallet groups by
common-input clustering;
CoinJoin excluded so
innocent owners never merge

3. CORRELATE

Network ⇔ chain fusion:
money-moved edges vs
controlled-by edges, with
country and operator



STACK — AND WHY EACH PIECE IS THERE

- **Python 3.10 · FastAPI · uvicorn** — one process serves the API, the workstation and the A4 reports
- **pandas · numpy · sklearn · networkx** — no GPU, no cloud, no model server
- **SQLite for state and records** — one file holds the analysis AND the audit trail; back it up whole
- **Vendored vis-network, Chart.js, 42 font files** — the graph and charts render with the cable unplugged
- **pytest · smoke task · offline check** — the working state is reproducible, not remembered

The map is the fusion made visible. 361 wallet groups and their network endpoints for one batch: 185 links, 55 of them control edges — an IP holding a wallet, which is a different fact from money moving.

Reproduce it: `python tasks.py smoke` generates a capture, trains from scratch, replays it and checks every endpoint. `python tasks.py offline` proves no shipped file reaches the network.

FEASIBILITY AND VIABILITY



TEAM VORTEX

Already running on the machine this was built on — the numbers below come from that run.

WHAT IS ALREADY FEASIBLE

Reads real exports.

A block-explorer style file with its own headers, satoshi amounts and junk rows is read and reported on in seconds.

Runs on one laptop.

The whole pipeline processes the demonstration capture in about 15 seconds per batch on CPU only.

Offline deployment.

One command starts it; Docker bakes the capture, the models and the first administrator into the image.

Scales by batches, not by hope.

A capture is replayed window by window, so the work is bounded by the day, not by the file.

Every output is checkable.

The report names the engine, the model hash and the capture hash that produced it.

RISKS, AND HOW THEY ARE HANDLED

A model that looks good offline and fails in use.

Trained on the scale it is served at; drift is measured per batch; the scorecard publishes the linear-model baseline beside it.

Garbage in, silent garbage out.

The gate reports what it set aside and why, before anything is scored.

A score read as a verdict.

Every screen, report and export carries 'a lead is not a finding of guilt', and the attribution reconciles to the number shown.

Network/chain mismatch: a file with no IP columns.

The analysis says so, drops the control edges, and reports what geography is lost.

Someone edits the record afterwards.

The audit log is append-only by database trigger; a capture replay clears the analysis and never the records.

MEASURED ON PLANTED TRUTH

97.8%

ROC-AUC, 4-fold grouped by batch

89.3% / 85.9%

precision / recall at the review floor

100%

precision in the top-25 leads

0.99

clustering agreement with true owners

0.033

Brier score — the probabilities are calibrated

44 / 44

planted entities detected, 40 in the first batch

IMPACT AND BENEFITS



What a cybercrime unit gets on the first afternoon, and what it costs them.

25,240

transactions read
4 daily batches

361

wallet groups resolved
content-derived identity

47

leads raised
each with its reason

183

changes between batches
what moved, and from what
to what

- Potential impact on the target audience
- Benefits of the solution (environmental, etc.)

WHO BENEFITS, AND HOW

NTRO analysts

A ranked queue instead of a graph they have to read from scratch — with the reason attached to every entry.

Law-enforcement officers.

An A4 lead report that names the evidence hash: something attachable to a file, not a screenshot.

Exchanges and banks.

Victim-notification sheets for poisoning campaigns, listing exactly who received a forged address.

Field teams.

The same queue on a phone, served by the local workstation, with no cloud account and no data leaving the room.

The institution itself.

An append-only record of who looked at what, so the tool survives an audit and a change of staff.

ECONOMIC AND OPERATIONAL CASE

No cloud bill, no GPU, no per-seat licence. It runs on the workstation the unit already owns, on Python 3.10 and a handful of pinned open-source libraries.

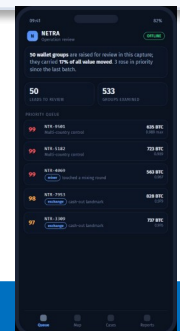
No data-protection exposure. Seized evidence and intelligence never leave the machine, because there is nowhere for them to go.

Recovered analyst time. Triage moves from reading a transaction graph to reading a ranked list of 47 entries out of 361 groups.

It keeps improving. Every decision an analyst records becomes a label; `python tasks.py retrain` trains a challenger that a human promotes — never one that promotes itself.

IN THE FIELD

The same queue, on a phone: the one-sentence answer, the two numbers that matter, and who to look at first — served locally, working offline.



RESEARCH AND REFERENCES



What this stands on, and where the claims in this deck can be checked.

Domain foundations

Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System.

Meiklejohn et al. (2013). A Fistful of Bitcoins: Characterizing Payments Among Men with No Names. IMC — the common-input ownership heuristic, and its limits.

Möser & Böhme (2015). Trends, Tips, Tolls: A Longitudinal Study of Bitcoin Transaction Fees. FC — value-flow tracing.

Bon & Shamir (2013). Quantitative Analysis of the Full Bitcoin Transaction Graph. FC — peer chains and mixing

Method and tooling

Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. JMLR.

Breiman, L. (2001). Random Forests. Machine Learning 45(1).

Liu, Ting & Zhou (2008). Isolation Forest. ICDM.

Saabas, T. (2014). Interpreting random forests — the decision-path attribution we use, chosen over SHAP because it reconciles exactly and runs in microseconds on an air-gapped host.

Hagberg, Schult & Swart (2006). Exploring Network Structure with NetworkX.

FastAPI, uvicorn, SQLite, vis-network, Chart.js — official documentation.

Standards and process

SIH 2026 Idea Submission template and evaluation criteria: novelty, complexity, clarity, feasibility, practicability, impact, UX (sih.gov.in).

NTRO problem statement SIH26146: AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic.

ISO/IEC 27037:2012 — handling of digital evidence; the reason the audit log is append-only and every output carries a hash.

• Details / Links of the reference and research work

Threat patterns used as detectors

FATF (2023). Targeted Update on Implementation of the FATF Standards on Virtual Assets and VASPs — typologies behind the risk features.

Chainalysis (2023–24). Crypto Crime Report — address poisoning and dusting as observed attacker behaviour.

Public exchange advisories on lookalike-address poisoning (prefix/suffix forgery) — the basis of our matched-character rule.

Everything claimed here is reproducible on the machine that produced this deck. `python tasks.py smoke` rebuilds a capture, trains the models, replays the batches and checks every endpoint; `python tasks.py accept <file>` reports exactly what was read from any capture you supply.